

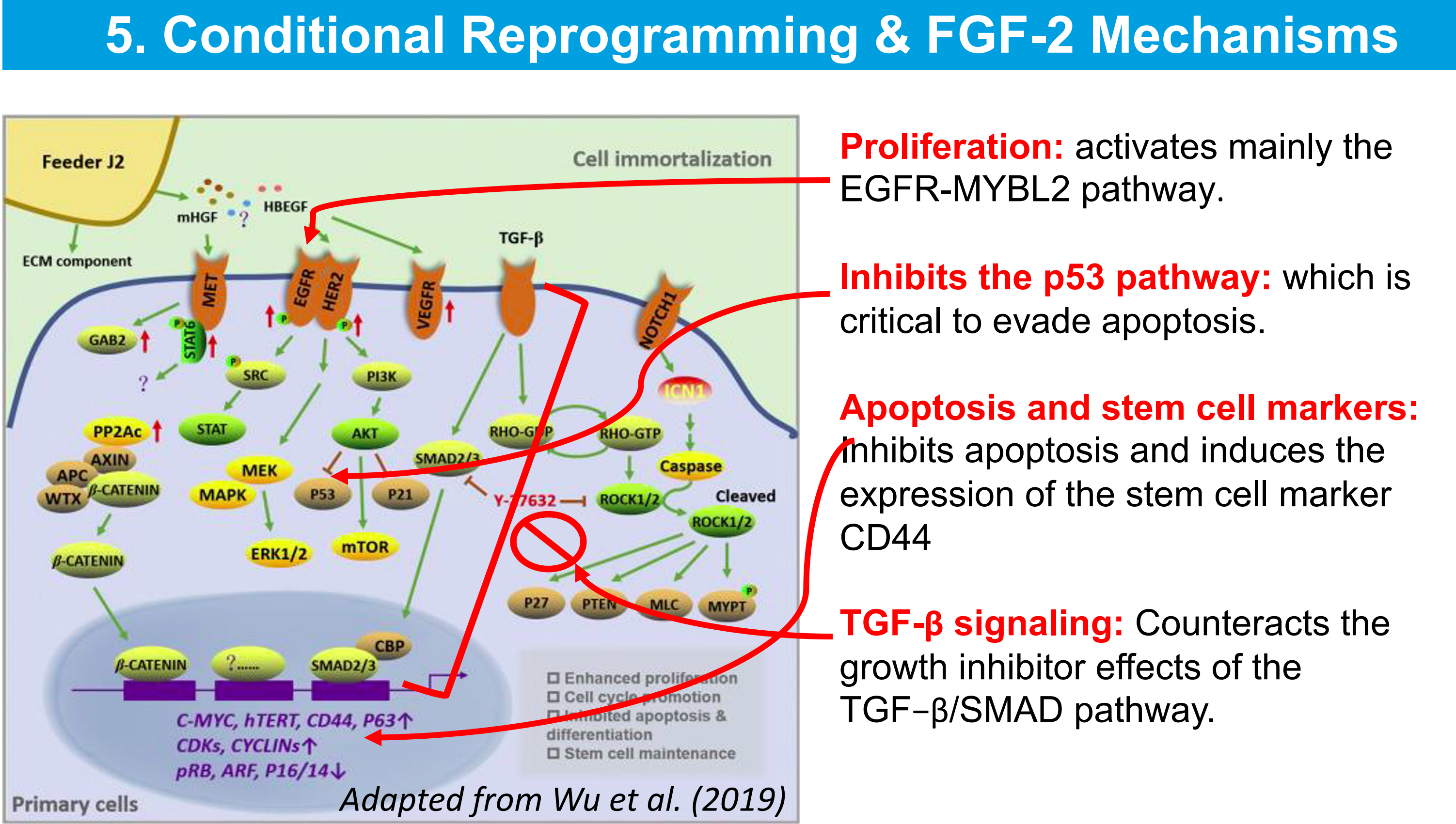
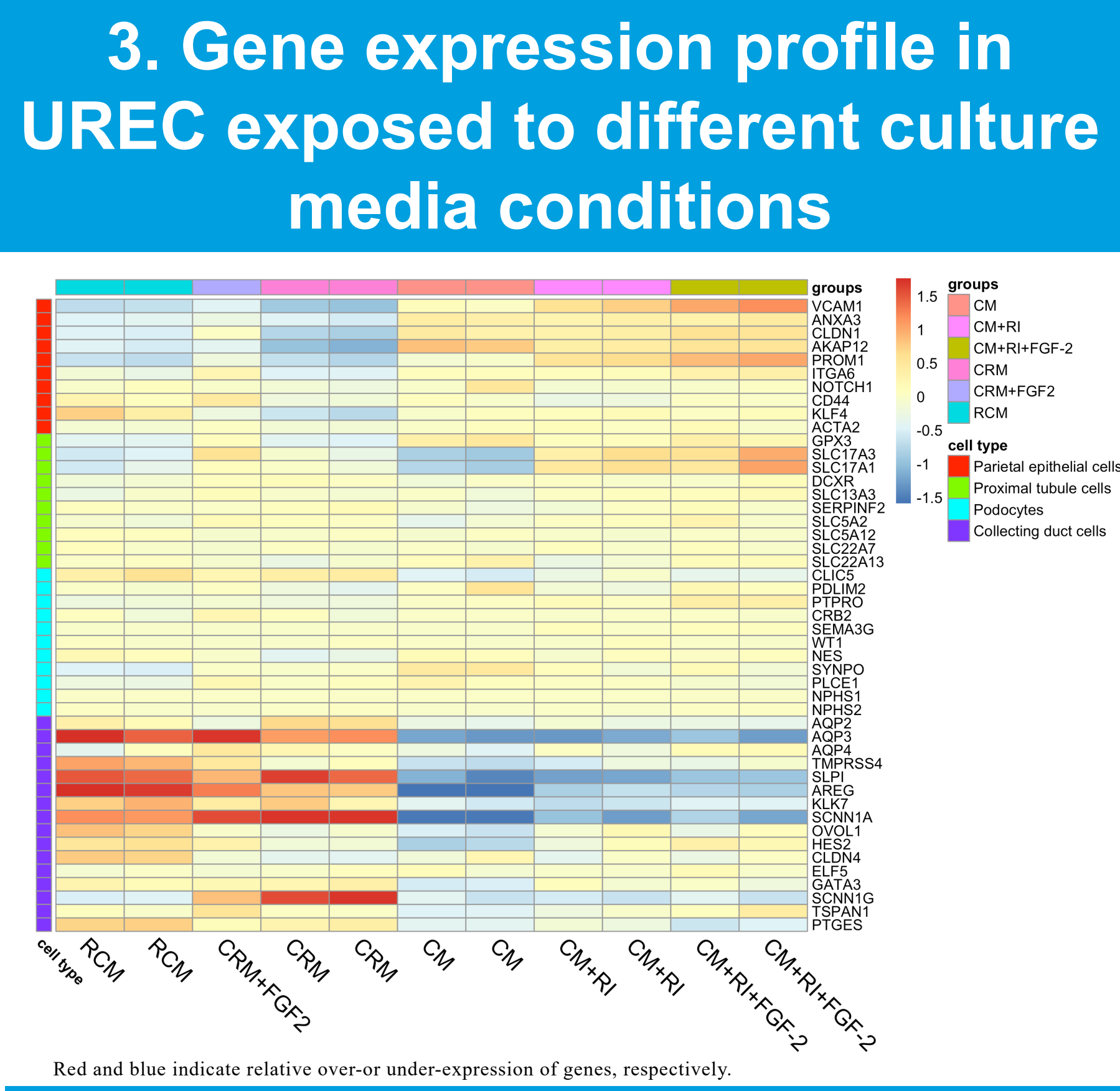
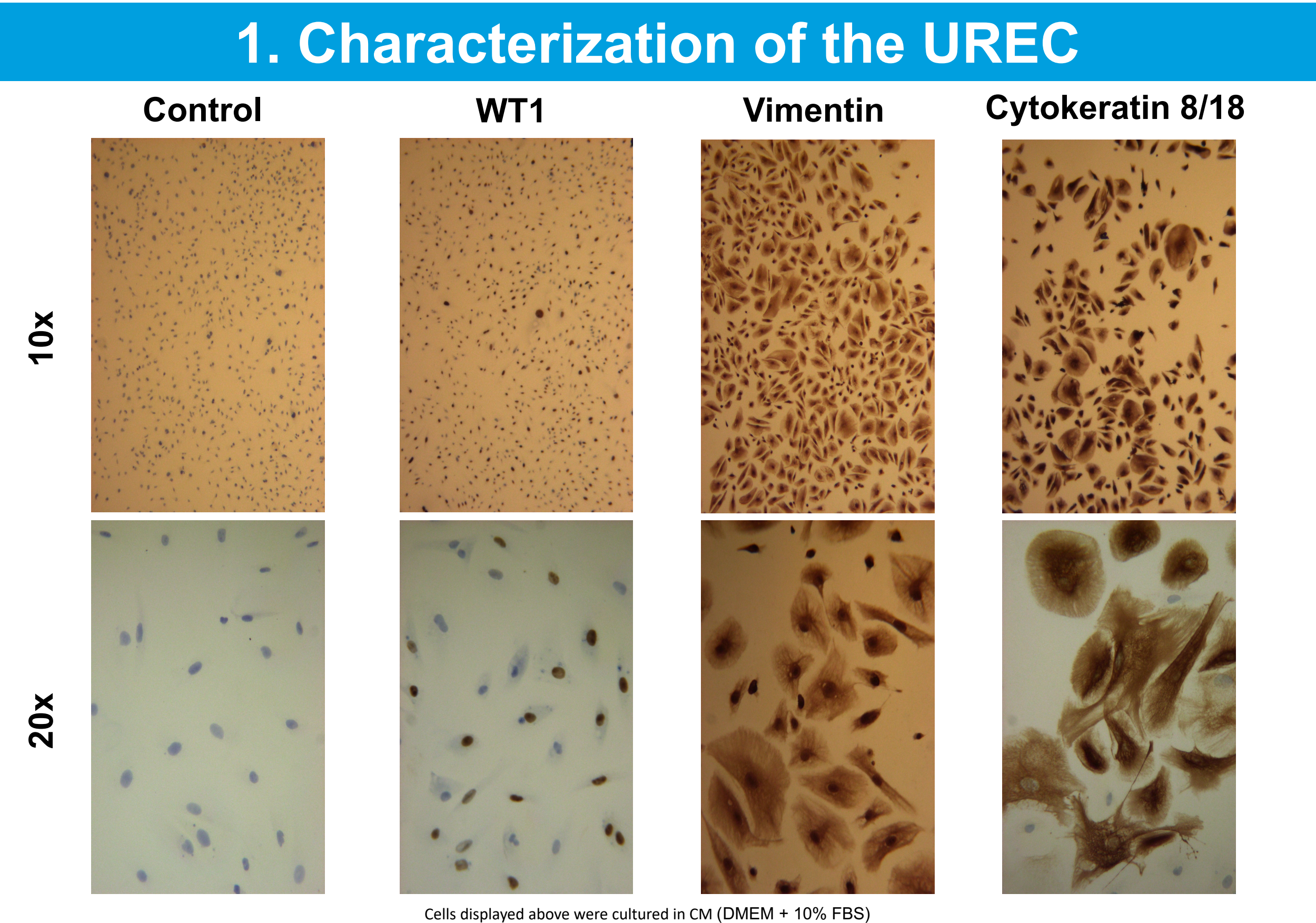
Discovery of Novel Tissue Culture Conditions to Expand the Growth of Primary Renal Epithelial Cells Cultured from the Urine of Children with Nephrotic Syndrome



Matthew Runyan¹, Pankaj Kumar², Jing Yu¹, and Patricio Ray¹
¹Child Health Research Center, Department of Pediatrics, University of Virginia, School of Medicine, Charlottesville, VA, USA ²Department of Biochemistry and Molecular Genetics, School of Medicine, University of Virginia, Charlottesville, VA, USA

Background

- One major limitation to study the pathogenesis of nephrotic syndrome (NS) in children is the lack of robust culture systems to expand primary podocytes and glomerular parietal epithelial cells (GPEC) derived from these patients.
- Previous studies showed that conditional reprogramming medium (CRM) allows the unlimited proliferation of primary epithelial cells derived from the breast, colon, lung, nose, prostate, without introducing genetic or viral manipulations.¹⁻³
- It is unknown whether a similar approach could facilitate the expansion of primary podocytes and GPEC cultured from the urine of children with NS.

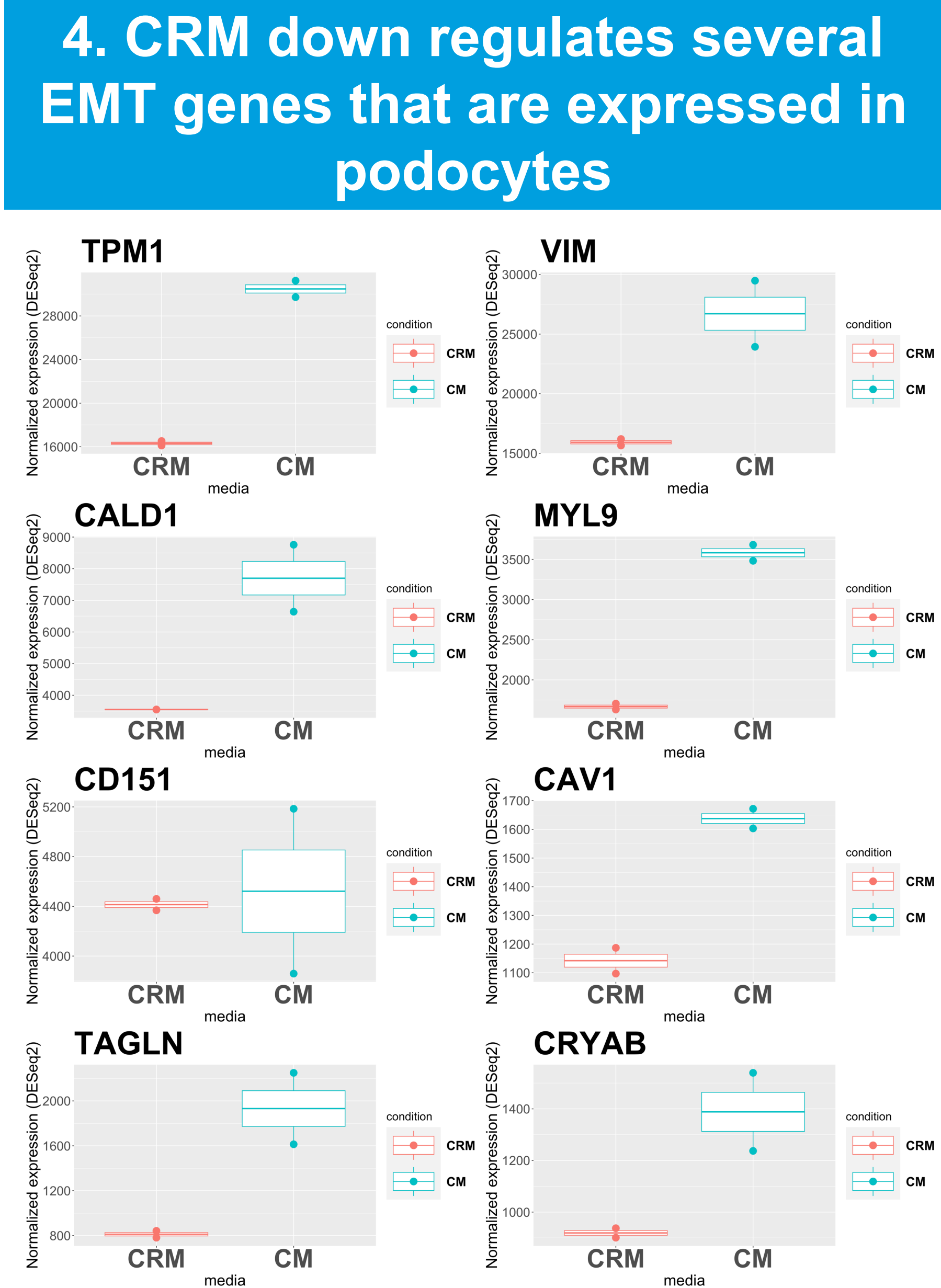
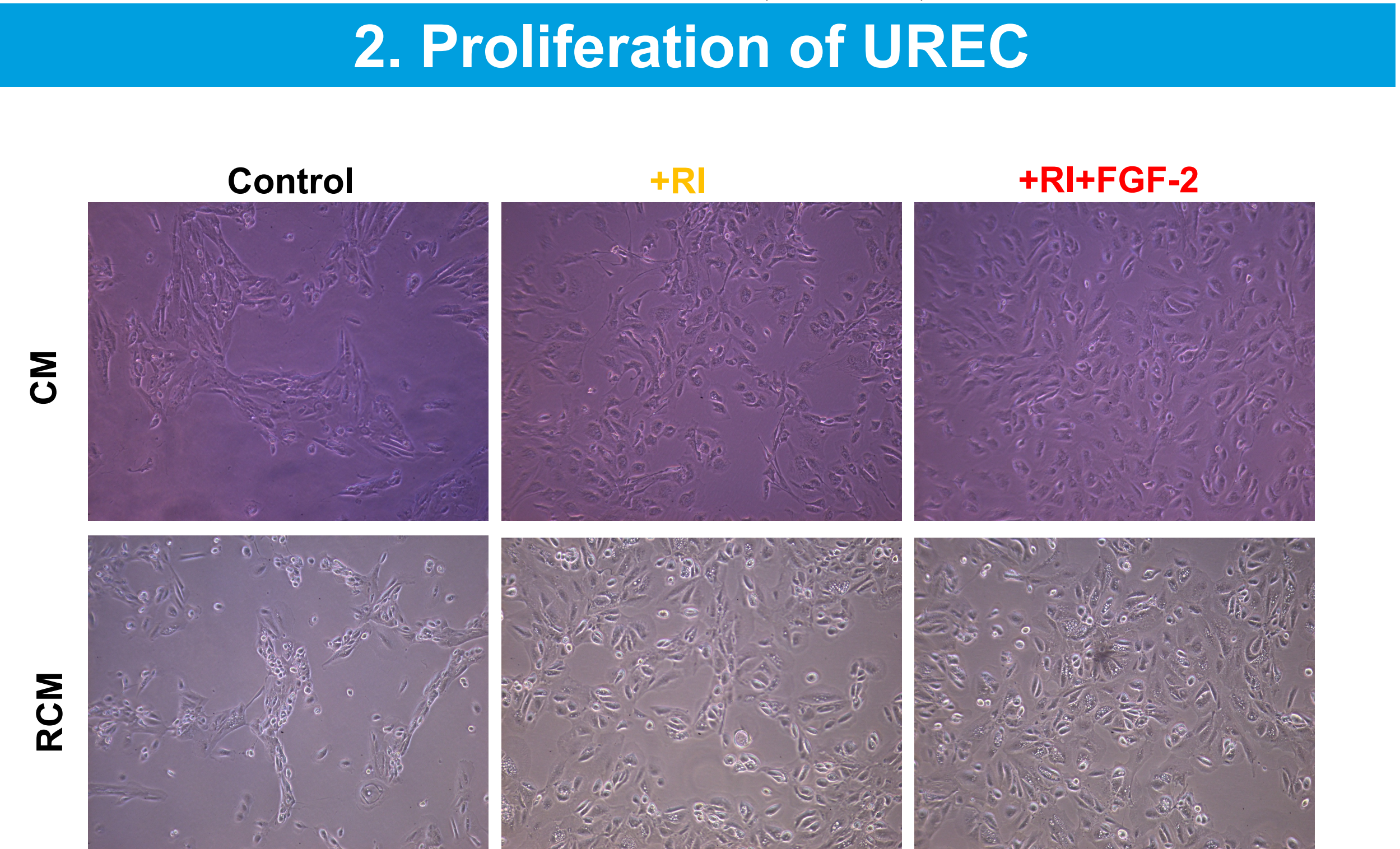


Objective

- To determine how CRM and/or other tissue culture conditions affect the proliferation, expansion, and bulk transcriptome profiling of primary urinary renal epithelial cells (UREC) cultured from a child with a primary NS.

Methods

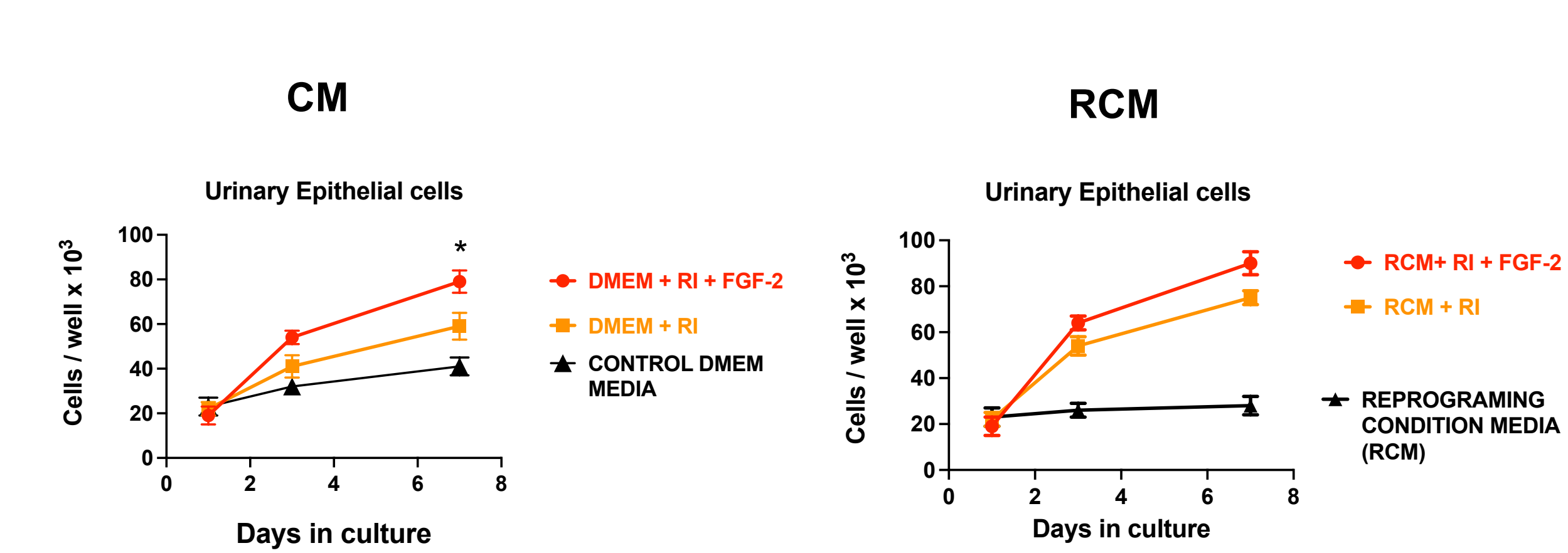
- UREC were isolated from the urine of a child with a primary NS, cultured in duplicates under the tissue culture conditions described below, processed for bulk transcriptome profiling, and analyzed as described in previous studies.¹⁻⁵



Conclusions

- Conditional Reprograming Media (CRM) affects the morphology, growth rate, and transcriptome profile of primary UREC cultured from a child with nephrotic syndrome.
- CRM downregulates the expression of EMT genes associated with podocytes, as well as the expression of genes associated with differentiated PEC and Podocytes.
- CRM induces the expression of genes associated with collecting duct epithelial cells.
- Control Media (CM = DMEM + 10% FBS) alone or in combination with FGF-2 maintains the growth rate and differentiation of podocytes and parietal epithelial cells.
- A significant number of parietal epithelial cells and podocytes are shed in the urine of children with NS and proliferate in the presence of FGF-2.
- Our findings are consistent with the results of previous studies showing that glomerular epithelial cells underlying the Bowman capsule may generate podocytes and proximal tubular cells⁶⁻⁷. CRM appears to facilitate this transition.

Culture Conditions	Abbreviation
Control Media (DMEM + 10% FBS)	CM
CM + Rock inhibitor Y-27632 (10 μM)	CM+RI
CM + RI + FGF-2 (10 ng/ml)	CM+RI+FGF2
Reprogramming Condition Medium (DMEM + condition medium from irradiated murine fibroblasts) ³	RCM
Conditional Reprograming Medium (RCM+ RI) ¹⁻³	CRM
CRM + FGF-2 (10 ng/ml)	CRM+FGF2



References

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Acknowledgements

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